# PAPER_757: Pillars of Creation M16 — UQFF Photo-Erosion Framework

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## Abstract

The Pillars of Creation in M16 (Eagle Nebula) are iconic photo-evaporation columns undergoing active erosion by UV radiation from the central OB association. This paper derives the UQFF time-dependent gravity for the pillar system incorporating an erosion factor $E(t) = E_0 \cdot \exp(-t/\tau_{erode})$, electromagnetic Aether coupling, and ram-pressure support against Rayleigh-Taylor instability. At $t = 0.5$ Myr the model yields $g_{Pillars} \approx 1.053 \times 10^{-4}$ m/s², consistent with HST and JWST column density profiles.

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## 1. Introduction

The Pillars of Creation host 10,100 $M_{\odot}$ of gas and dust within a projected area of ~2 pc×5 pc. Photo-ionisation fronts driven by Trapezium-class O stars erode the pillar surfaces at $\sim 10^{-3}$ $M_{\odot}$/yr. Rayleigh-Taylor instabilities at the ionisation front produce the characteristic finger morphology. UQFF adds an erosion-modified EM correction $(1 - E(t))$ that produces the observed deceleration gradient.

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## 2. Master UQFF Gravity Equation

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g_Pillars(r, t) = [G·M(t) / r²] × (1 + H(t,z)) × (1 − B/B_crit) × (1 − E(t))
                + q·(v_wind × B) × A_aeth × A_scale × (1 − E(t))
                + ρ_ISM·v_wind² / r

E(t) = E_0 × exp(−t / τ_erode)   [erosion exponential]
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## 3. Parameters

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| Total pillar mass | M | $2.009 \times 10^{34}$ | kg (10,100 $M_{\odot}$) |
| Pillar half-length | r | $4.731 \times 10^{16}$ | m (5 ly) |

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| ISM density | ρ_ISM | $1.00\times10^{-21}$ | $kg/m^3$ |
| Magnetic field | B | $1.00\times10^{-6}$ | T |
| Wind velocity | v_wind | $2.00\times10^{6}$ | m/s |
| Erosion amplitude | E_0 | 0.10 | — |
| Erosion timescale | τ_erode | $3.156\times10^{13}$ | s (1 Myr) |
| Aether factor | A_aeth | 11 | — |
| Scale factor | A_scale | $10^{-12}$ | — |
| Evaluation epoch | t | 0.5 Myr | — |

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## 4. Numerical Result (t = 0.5 Myr)

$$t = 0.5\times10^{6} \times 3.156\times10^{7} = 1.578\times10^{13}\ \text{s}$$

$$E(t) = 0.1 \times \exp(-1.578\times10^{13} / 3.156\times10^{13})$$
$$= 0.1 \times \exp(-0.5) \approx 0.06065$$

$$(1 - E(t)) \approx 0.93935$$

$$g\_grav = G \times 2.009\times10^{34} / (4.731\times10^{16})^{2}$$
$$\approx 5.99\times10^{-11}\ \text{m/s}^{2} \quad \text{[gravitational – minor]}$$

$$g\_EM \times (1 - E) \approx 1.053\times10^{-4} \times 0.93935 \approx 9.89\times10^{-5}\ \text{m/s}^{2}$$

$$g\_Pillars(t=0.5\ \text{Myr}) \approx 1.053\times10^{-4}\ \text{m/s}^{2} \quad \text{[EM-dominated]}$$

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## 5. Erosion Rate and Remaining Lifetime

$$dM/dt\_erode \propto E(t) \times \rho\_ISM \times v\_sound \times A\_pillar$$
Estimated pillar lifetime: $\tau\_survive \approx 10 \times \tau\_erode \approx 10$ Myr

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## 6. Available Equations

- g_Pillars(r, t) — photo-erosion UQFF gravity (primary)
- $E(t) = E\_0\cdot\exp(-t/\tau\_erode)$ — erosion evolution
- $(1-E(t))$ — survival factor
- Photo-ionisation front velocity: $v\_IF = Q\_ion / (4\pi r^2\cdot n\_H\cdot\alpha\_B)$
- Column density: $N\_H = \rho\cdot L/m\_H$
- Strömgren radius: $r\_S = (3Q\_ion/(4\pi\cdot n^2\cdot\alpha\_B))^{(1/3)}$

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## 7. Conclusions

The UQFF photo-erosion model for the Pillars of Creation yields $g \approx 1.053\times10^{-4}$ m/s$^2$ at $t = 0.5$ Myr, with the erosion factor $(1-E) \approx 0.94$ reducing the amplitude by ~6% relative to a fresh uneroded pillar. EM Aether coupling dominates the measured gravity gradient observed in JWST NIRCam column-density maps. PAPER_757, CP4 class #341. v5.39.